

ROHHITH GUNASEKARAN

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PROFESSIONAL SUMMARY

Engineering student focused on system-level problem solving, backend architecture, and AI-driven platforms. Experienced in designing simulation engines, distributed systems, and real-time applications. Strong in applying AI, optimization, and system design principles to build scalable, production-oriented systems.

CORE SKILLS



PROGRAMMING

Python, C, C++, JavaScript



BACKEND & SYSTEMS

FastAPI, REST APIs, WebSockets, System Design



FRONTEND

React, HTML, CSS



DATABASES

PostgreSQL, MySQL, MongoDB, SQLite



AI & ML

Computer Vision, AI Pipelines, Model Training, Intelligent Agents



TOOLS & PLATFORMS

Git, Docker (Basic), Linux (Basic), Cloud (AWS/Azure Exposure)



CONCEPTS

Data Structures, Algorithms, OOP, DBMS, Operating Systems, Networking, Cyber Security Basics

EXPERIENCE



Project Development Cell — Internship (Ongoing)

Coimbatore Institute of Technology

- Selected to develop a system-level project using institutional GPU infrastructure and lab facilities.
- Working on a decentralized compute and GPU virtualization platform involving real-time streaming, virtualization, and resource management.



Cybersecurity Internship (Winter)

CICT – Product Development Cell

- Completed internship focusing on cybersecurity concepts, vulnerabilities, and system analysis.
- Gained exposure to security frameworks and threat modeling.

ACHIEVEMENTS



VIT Hackathon – Finalist

- Reached the final stage with a system-level project under competitive evaluation.

EDUCATION



B.E. Computer Science Engineering

Coimbatore Institute of Technology (Anna University)

IN-PROGRESS PROJECT



PeerLink — Decentralized GPU Virtualization & Remote Execution Platform

[In Progress]

Tech Direction: Docker/KVM, WebRTC, WebSockets, FFmpeg, CUDA

- Designing peer-to-peer system for sharing GPU, CPU, and memory across trusted nodes.
- Implements containerized execution with resource isolation (cgroups, namespaces).
- Enables real-time remote execution via low-latency streaming using WebRTC and FFmpeg.
- Focused on decentralized compute to eliminate dependency on cloud GPU infrastructure and enable scalable, accessible high-performance computing.

FEATURED PROJECTS (TOP 3)

1	AttackSim — AI-Driven Attack Path Simulation Engine <i>Tech: Python, Graph Algorithms, Monte Carlo, ML, Next.js</i> - Modeled infrastructure as directed graphs for attack path computation - Built multi-stage attack chains (initial access, lateral movement, privilege escalation, exfiltration) - Applied Monte Carlo simulation for probabilistic risk evaluation - Integrated ML exploit prediction and MITRE ATT&CK-based modeling	Simulates real-world attack paths across infrastructure using graph modeling, multi-stage kill chains, and probabilistic risk scoring. Integrates ML models and MITRE ATT&CK for advanced threat analysis.
2	DocVerify — Blockchain-Based Document Verification System <i>Tech: React, Node.js, PostgreSQL, Redis, Ethereum, IPFS</i> - Designed SHA-256 based hashing pipeline for deterministic fingerprinting - Integrated IPFS for decentralized storage and Ethereum smart contracts for immutability - Built scalable backend with async processing using queues (BullMQ) - Ensures proof-based validation replacing traditional trust-based verification	End-to-end document verification system using cryptographic proof, blockchain anchoring, and decentralized storage to ensure tamper-proof and deterministic validation.
3	ResiliAI — AI Infrastructure Recovery System <i>Tech: Python, Simulation Systems, LLM Integration, Hugging Face</i> - Built state-driven simulation environment for distributed system failures - Developed hybrid decision engine (rule-based + adaptive control) - Modeled latency, load, and failures to trigger intelligent recovery actions - Deployed reproducible system with external evaluation pipeline	Simulates infrastructure failures and autonomously applies recovery actions using a hybrid decision engine. Optimizes system resilience under real-world stress conditions.

OTHER PROJECTS

PackAI — AI-Based Package Inspection System <i>Tech: Python, YOLOv8, CNN, ViT, Azure</i>	AI-powered system for package damage detection and classification using multi-stage computer vision models. Generates accept/reject decisions with tamper-proof evidence logging.
PulseLink — Real-Time Alert System <i>Tech: React, FastAPI, WebSockets, PostgreSQL</i>	Real-time campus alert and communication platform with WebSocket-based delivery, role-based access, and acknowledgement tracking for accountability.
VentureSim — Decision Intelligence Platform <i>Tech: Python, System Modeling</i>	Decision engine for evaluating startup ideas. Analyzes market viability, financial sustainability, risk factors, and simulates growth trajectories to generate strategic insights.
macOS Web Simulation <i>Tech: HTML, CSS, JavaScript (Vanilla)</i>	Browser-based macOS simulation with boot sequence, window management, system processes, and persistent state. Built entirely without frameworks.
Zephyx — CLI Issue Resolution System <i>Tech: Python, MySQL</i>	Console-based issue tracking and resolution system with role-based access, priority scoring, workload-based assignment, and analytical reporting in real-time.

TECHNICAL EXPERTISE

System Design & Architecture: Simulation Engines, Distributed Systems, Graph Modeling, Decision Systems, Optimization

AI & Machine Learning: Computer Vision, AI Pipelines, Model Training, Intelligent Agents, Hybrid Decision Engines

Backend Engineering: API Design, Async Processing, WebSockets, Authentication, System Architecture

Databases: SQL, NoSQL, Schema Design, Data Modeling, Query Optimization

Infrastructure & Security: Blockchain, Cryptography, System Security, Risk Modeling, Threat Analysis

Tools & Platforms: Git, Linux, Docker (Basic), AWS/Azure (Exposure)

CORE COMPETENCIES

	Problem Solving & Analytical Thinking
	System Design & Architecture
	AI-Driven Development
	Security & Risk Analysis
	Performance Optimization
	Ownership & Execution